Data Science Program

Capstone Report - Spring 2026

Predicting ALS Disease Progression Using Longitudinal ALSFRS-R Data

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A capstone presented in fulfillment of the requirements for the degree of Bachelor's of Science in Data Science

supervised by
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Preface

I am an undergraduate data science student interested in applying machine learning to real-world problems where data can support better decision making. As a former Biology major, I had the opportunity to conduct research in a neurodegenerative disease lab at Ursinus College, where I became interested in how computational tools can help study diseases with complex progression patterns. I chose this project because amyotrophic lateral sclerosis, or ALS, is a serious neurodegenerative disease with progression patterns that differ widely across patients. This made ALS progression prediction a meaningful research setting for practicing the full data science workflow, including data cleaning, longitudinal feature engineering, model evaluation, and interpretation. The intended audience for this report includes data science students, healthcare researchers, and clinicians who are interested in how structured clinical data can be used to estimate disease progression while still recognizing the limitations of predictive models in healthcare.

Acknowledgements

I would like to thank Prof. Burnett for her guidance throughout DATS 4001 and for helping structure the capstone around the practice of reproducible, real-world data science. I would also like to thank the GW Data Science Department for supporting this capstone project and providing a platform to develop and present this work.

Abstract

Amyotrophic lateral sclerosis, or ALS, is a progressive disease where functional decline differs substantially across patients, making future disease severity difficult to estimate from a single clinical visit. This problem is interesting because ALSFRS-R scores provide repeated measurements of patient function, but the data are longitudinal, incomplete, and heterogeneous across individuals. I developed a reproducible Python modeling workflow that cleans longitudinal ALS clinical records, engineers patient-history features, compares baseline and machine learning models, and evaluates prediction error across disease severity groups. The best model achieved approximately 1.81 mean absolute error, 2.57 root mean squared error, and an R^2 near 0.91, showing reasonably strong overall prediction while also revealing weaker reliability for lower-functioning patients.

Keywords

ALS; ALSFRS-R; machine learning; clinical prediction; data science; Capstone; GWU

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1 Introduction

Amyotrophic lateral sclerosis, often abbreviated ALS, is a progressive neurodegenerative disease that affects motor neurons and gradually reduces a person's ability to perform physical functions. Because the disease affects patients differently, predicting future functional decline is difficult. Two patients can have similar function at one visit but follow very different trajectories over the next several months or years. This makes ALS progression a useful but challenging problem for longitudinal data science.

The main research question for this project is whether longitudinal patient history can be used to predict future ALS functional severity. I focus on the ALS Functional Rating Scale Revised, or ALSFRS-R, as the main outcome variable. ALSFRS-R is a clinical scale used to measure functional status in ALS, with total scores ranging from 0 to 48. Higher scores indicate better function, while lower scores indicate more severe impairment. In this report, predicting ALS progression means predicting ALSFRS-R total score over time rather than diagnosing ALS or predicting every biological feature of the disease.

The goal of this capstone was to build a reproducible Python workflow for modeling ALSFRS-R progression from longitudinal clinical data. The final workflow uses PRO-ACT as the main modeling source and Target ALS as supplementary clinical context. I cleaned repeated patient visit records, preserved the longitudinal structure of the data, engineered features that describe each patient's clinical history, trained multiple regression models, and evaluated model performance using mean absolute error, root mean squared error, and R^2.

This project produced several main findings. First, longitudinal patient history is highly predictive of future ALSFRS-R score. In particular, the previous ALSFRS-R score was one of the strongest predictors of the next observed score. Second, learned machine learning models improved over a naive previous-visit baseline, but the improvement was modest. This suggests that the strongest signal comes from patient history itself, not only from more complex model architecture. Third, prediction error was not uniform across patient groups. The model had the highest error for lower-functioning patients, which means the model was less reliable for patients with more advanced functional impairment.

The project is best understood as a reproducible modeling pipeline and proof of concept rather than a standalone clinical decision tool. A model that explains a large share of variation across a dataset can still be uncertain for a specific patient. For that reason, the report emphasizes not only overall model metrics, but also the real-world meaning of error and the importance of subgroup analysis.

2 Related Work

ALSFRS-R has been widely used as a clinical measure of disease severity and functional decline in ALS. Cedarbaum et al. introduced the revised ALS Functional Rating Scale as a way to incorporate respiratory function and provide a standardized measure of daily functional ability in ALS patients [1]. My project

builds directly on this measurement framework by using ALSFRS-R total score as the prediction target. Unlike the original scale development work, this capstone does not evaluate the clinical design of ALSFRS-R itself. Instead, it treats the score as an established longitudinal outcome that can be modeled using data science methods.

The PRO-ACT database is one of the most important data resources for ALS modeling because it brings together de-identified clinical trial data into a structured research dataset. Atassi et al. describe the design and early analysis of PRO-ACT, including its use for identifying predictive features in ALS clinical data [2]. My project uses PRO-ACT in a narrower way. Rather than building a large clinical risk model across all available variables, I focus on constructing a reproducible undergraduate-level pipeline that moves from raw longitudinal visit records to model-ready features, predictions, and interpretation.

Prior work has also attempted to predict ALS functional decline and survival using clinical data. Ong et al. developed models for predicting functional decline and survival in ALS, showing that structured patient information can support predictive modeling of disease outcomes [3]. That work is more extensive than this capstone in terms of clinical modeling scope. My project is more focused on the data science workflow itself. It compares several standard regression approaches and emphasizes interpretability through error analysis, especially by disease severity group.

Longitudinal ALS research also shows that patient outcomes are influenced by many clinical and demographic factors. Rooney et al. studied survival patterns among ALS patients and demonstrated that ALS outcomes vary substantially across patient groups [4]. Su et al. conducted a large meta-analysis of survival predictors and showed that ALS prognosis cannot be reduced to one simple factor [5]. These studies support the motivation for including patient-level features beyond time alone. My project follows the same general idea, but uses a narrower set of available structured variables and focuses on ALSFRS-R score prediction rather than survival.

ALS is also biologically heterogeneous. Tam et al. identified molecular subtypes of ALS using post-mortem cortex samples, which supports the idea that patients may differ not only clinically but also biologically [6]. This is relevant to the limitations of my project. The final model mostly relies on structured clinical visit variables. It does not fully incorporate genetic, biomarker, or molecular subtype data. As a result, the model can learn from observed patient history, but it may miss biological factors that help explain why some patients decline more quickly than others.

Together, these sources show that ALSFRS-R is an established outcome measure and ALS progression is shaped by clinical and biological heterogeneity. My work builds on these ideas by creating a reproducible Python pipeline for longitudinal ALSFRS-R prediction and by connecting model error back to patient severity. The main contribution is not a new clinical scale or a new machine learning algorithm. The contribution is an end-to-end data science workflow that shows how far standard machine learning models can go with longitudinal clinical history, and where their reliability begins to weaken.

3 Solution

The solution developed for this project is a reproducible Python modeling workflow for predicting ALSFRS-R total score from longitudinal patient data. The workflow follows the main stages of a data science pipeline: data acquisition, cleaning, exploratory analysis, feature engineering, model training, evaluation, and interpretation. Each step was designed to preserve the longitudinal structure of the data because ALS progression is a time-based problem.

3.1 Data Sources and Data Structure

The primary modeling dataset was PRO-ACT, a de-identified ALS clinical dataset with repeated patient visit information. The broader project also considered Target ALS as supplementary context, but the final modeling workflow was centered on PRO-ACT because it provided the strongest structured longitudinal source for ALSFRS-R measurements. The key variables used in the modeling pipeline included patient identifiers, visit timing, ALSFRS-R total score, demographic variables, and onset-related fields.

A key decision was to avoid collapsing the dataset into one row per patient. A single-row patient summary would remove the visit-to-visit structure that makes progression modeling possible. Instead, each row in the modeling table represented a patient visit. This allowed the model to learn from repeated observations over time and from features that described each patient's previous clinical history.

3.2 Exploratory Analysis

Before modeling, I examined the distribution of ALSFRS-R scores and the shape of patient trajectories over time. Figure 1 shows a sample of patient ALSFRS-R trajectories. The x-axis is measured in days since the baseline visit, and the y-axis is the ALSFRS-R total score. The figure shows that patients do not decline at the same rate. Some trajectories drop quickly, while others decline more gradually or remain relatively stable for longer periods. This supports the need for patient-level longitudinal modeling rather than a single average decline curve.

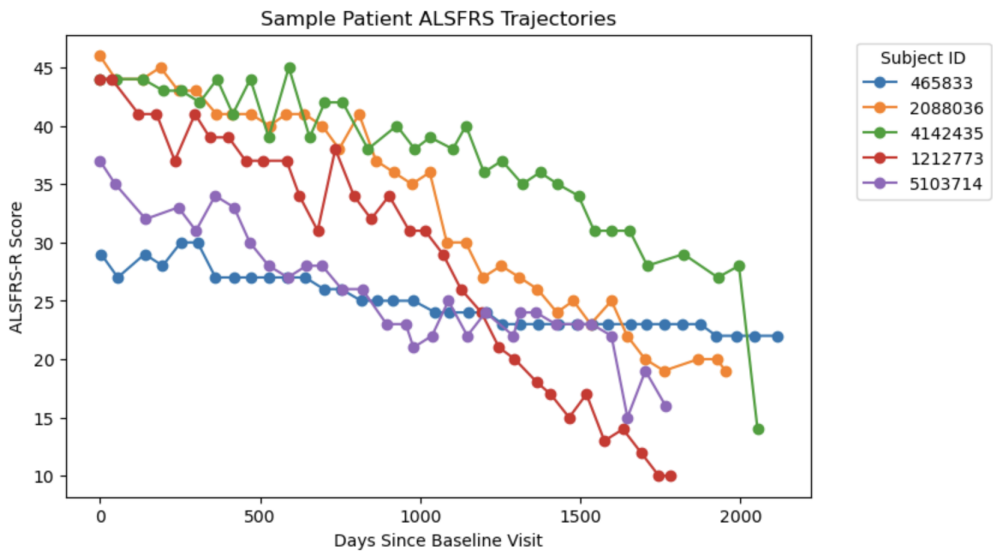

Figure 1: Sample patient ALSFRS-R trajectories over time. The x-axis shows days since baseline visit, and the y-axis shows ALSFRS-R total score. The variation across patient trajectories shows why longitudinal patient history is important for modeling ALS progression.

Figure 2 shows the distribution of ALSFRS-R total scores in the modeling dataset. Most observations are concentrated in the moderate-to-high functional range, while very low scores appear less frequently. This matters because the distribution of the target variable affects what the model has the most opportunity to learn. If the model sees more examples of moderate and high function than low function, it may be better trained for those ranges. This point becomes important later when analyzing prediction error by severity group.

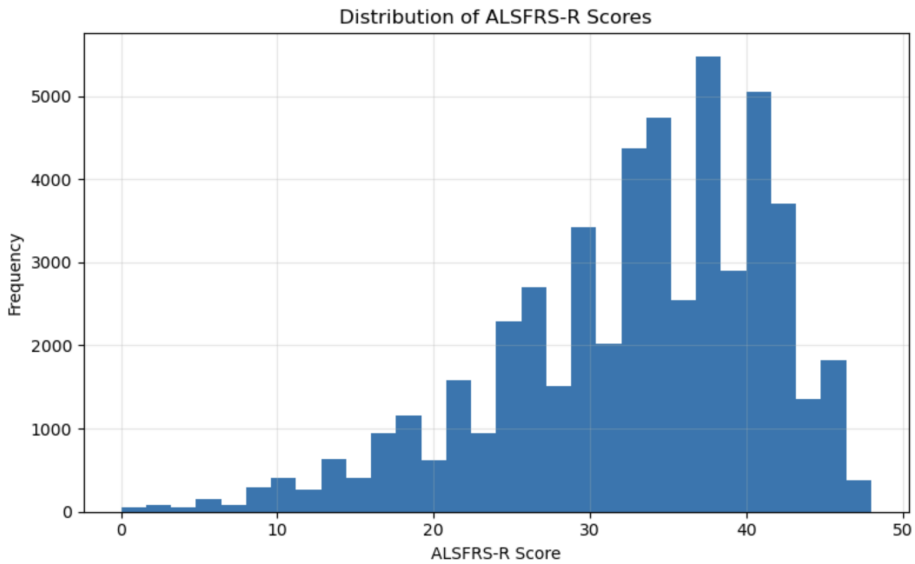

Figure 2: Distribution of ALSFRS-R total scores in the modeling dataset. Most observations fall in the moderate-to-high functional range, with fewer observations at very low scores. This distribution provides context for later differences in model error across severity groups.

3.3 Data Cleaning and Preparation

The preprocessing stage focused on creating a clean and consistent longitudinal modeling table. Column names were standardized so that key variables could be used consistently across notebooks and scripts. Records with missing or unusable core values were handled carefully because the prediction task depended on valid patient identifiers, visit timing, and ALSFRS-R scores. The key standardized variables included `subject_id`, `alsfrs_delta`, and `alsfrs_r_total`.

The variable `alsfrs_delta` was used as the main time variable. It represents the number of days since a patient's baseline visit. For example, a baseline visit has a value near day 0, while a follow-up visit 180 days later has an `alsfrs_delta` value of about 180. This variable allowed the workflow to place each visit in time relative to the patient's starting point.

3.4 Feature Engineering

Feature engineering was the most important step in turning clinical records into a usable prediction dataset. Since the goal was to predict ALSFRS-R score over time, the model needed features that represented both current visit timing and prior patient history.

The main engineered variables included visit number, baseline ALSFRS-R score, previous ALSFRS-R score, time since first observed visit, and change from the previous visit. The previous ALSFRS-R score was especially important because recent functional status naturally contains information about near-future functional status. Baseline ALSFRS-R score provided context about the patient's starting severity. Time-based variables helped the model account for how far each visit occurred from baseline and how much time had passed between observations.

Table 1 summarizes the main modeling features and their purpose.

Feature	Description	Purpose
`alsfrs_r_total`	ALSFRS-R total score	Prediction target
`alsfrs_delta`	Days since baseline visit	Captures progression time
`prev_alsfrs_r_total`	Previous ALSFRS-R score	Captures recent patient status
`baseline_alsfrs_r_total`	Baseline ALSFRS-R score	Captures initial severity
`visit_number`	Ordered visit count	Represents repeated observations
`delta_from_prev_visit`	Days since previous visit	Captures visit spacing
`time_since_first_visit`	Cumulative follow-up time	Represents patient history
Age and sex	Demographic variables	Adds patient context
Onset variables	Onset timing and site	Adds clinical context

Table 1: Key modeling features used in the ALSFRS-R prediction pipeline. Time variables are measured in days where applicable, and ALSFRS-R variables are measured by score points.

The main reason for using these features was to encode disease history. A model that only sees a current visit has limited context. A model that also sees the previous score, baseline score, and time between visits can better distinguish between a patient who is stable and a patient who is declining quickly.

3.5 Modeling Choices

I compared four modeling approaches: a naive previous-visit baseline, linear regression, random forest regression, and gradient boosting regression. These models were selected because they represent different levels of complexity and interpretability.

The naive previous-visit baseline predicted a patient's current ALSFRS-R score using the previous observed ALSFRS-R score. This baseline was important because ALSFRS-R values are usually strongly related from one visit to the next. If a machine learning model cannot improve over this simple baseline, then it may not be adding much value.

Linear regression was included as a simple and interpretable learned model. It assumes a mostly linear relationship between the features and the target score. This makes it easier to explain, but it may miss nonlinear patterns in disease progression.

Random forest regression was included because it can capture nonlinear relationships and interactions between variables. For example, the effect of time since baseline may depend on the patient's previous score or starting severity. A random forest can model these kinds of interactions without requiring them to be manually specified.

Gradient boosting regression was included because it often performs well on structured tabular data. It builds decision trees sequentially, with each new tree attempting to reduce errors made by earlier trees. This makes it a strong candidate for predictive performance, although it can be less straightforward to explain than linear regression.

3.6 Evaluation Strategy

The models were evaluated using mean absolute error, root mean squared error, and R^2. Mean absolute error, or MAE, measures the average absolute prediction error in ALSFRS-R score. Root mean squared error, or RMSE, also measures prediction error but gives more weight to larger mistakes. The R^2 score measures how much of the variation in ALSFRS-R scores is explained by the model across the test data.

These metrics were useful for different reasons. MAE was the most interpretable because it directly describes error in ALSFRS-R score. RMSE helped identify whether some predictions had larger misses. R^2 gave a dataset-level measure of how well the model captured overall variation. I also evaluated error by severity group because one overall score can hide differences in reliability across patient types.

4 Results and Discussion

4.1 Experimentation Protocol

After preprocessing and feature engineering, the modeling dataset was split into training and test sets. The models were fit on the training data and evaluated on held-out test data. This setup allowed the evaluation to measure how well the models generalized to observations not directly used for fitting.

The naive previous-visit baseline served as the main benchmark. This was a practical baseline because recent ALSFRS-R score is already expected to be highly predictive of the next observed score. The learned models were then compared against this baseline using MAE, RMSE, and R^2. For MAE and RMSE, lower values indicate better performance. For R^2, higher values indicate that the model explains more variation in the target score.

4.2 Model Comparison

Table 2 summarizes the model performance results. The naive previous-visit baseline achieved an MAE of about 1.94, an RMSE of about 3.01, and an R^2 of about 0.88. Linear regression improved on that baseline, with an MAE of about 1.83 and an R^2 of about 0.91. Random forest and gradient boosting performed slightly better, with gradient boosting producing the strongest overall metrics.

Model	MAE	RMSE	R^2
Naive previous-visit baseline	1.94	3.01	0.88
Linear regression	1.83	2.62	0.91
Random forest	1.82	2.60	0.912
Gradient boosting	1.81	2.57	0.914

Table 2: Model performance comparison on the held-out test set. MAE and RMSE are measured by ALSFRS-R score.

The table shows that machine learning models improved over the naive baseline, but the improvement was modest. This is an important result. It suggests that previous ALSFRS-R score already carries a large amount of predictive signal. The learned models add value by combining previous score with timing, baseline severity, visit spacing, and patient-level variables, but they do not completely change the prediction problem. In other words, the data signal matters more than model complexity alone.

The differences between the learned models are also meaningful. Linear regression is easier to interpret but is limited by its linear assumptions. Random forest can capture nonlinear relationships and feature interactions. Gradient boosting can sequentially correct earlier prediction errors, which helps explain why it performed slightly best. Still, the small difference between random forest and gradient boosting suggests that the major modeling gain came from using longitudinal patient history, not from choosing the most complex model.

4.3 Actual Versus Predicted Scores

Figure 3 shows actual versus predicted ALSFRS-R total scores. The dashed diagonal line represents perfect prediction. Points near the line indicate that the predicted score was close to the actual score. The points generally follow the diagonal pattern, which shows that the model captures the broad relationship between patient history and ALSFRS-R score.

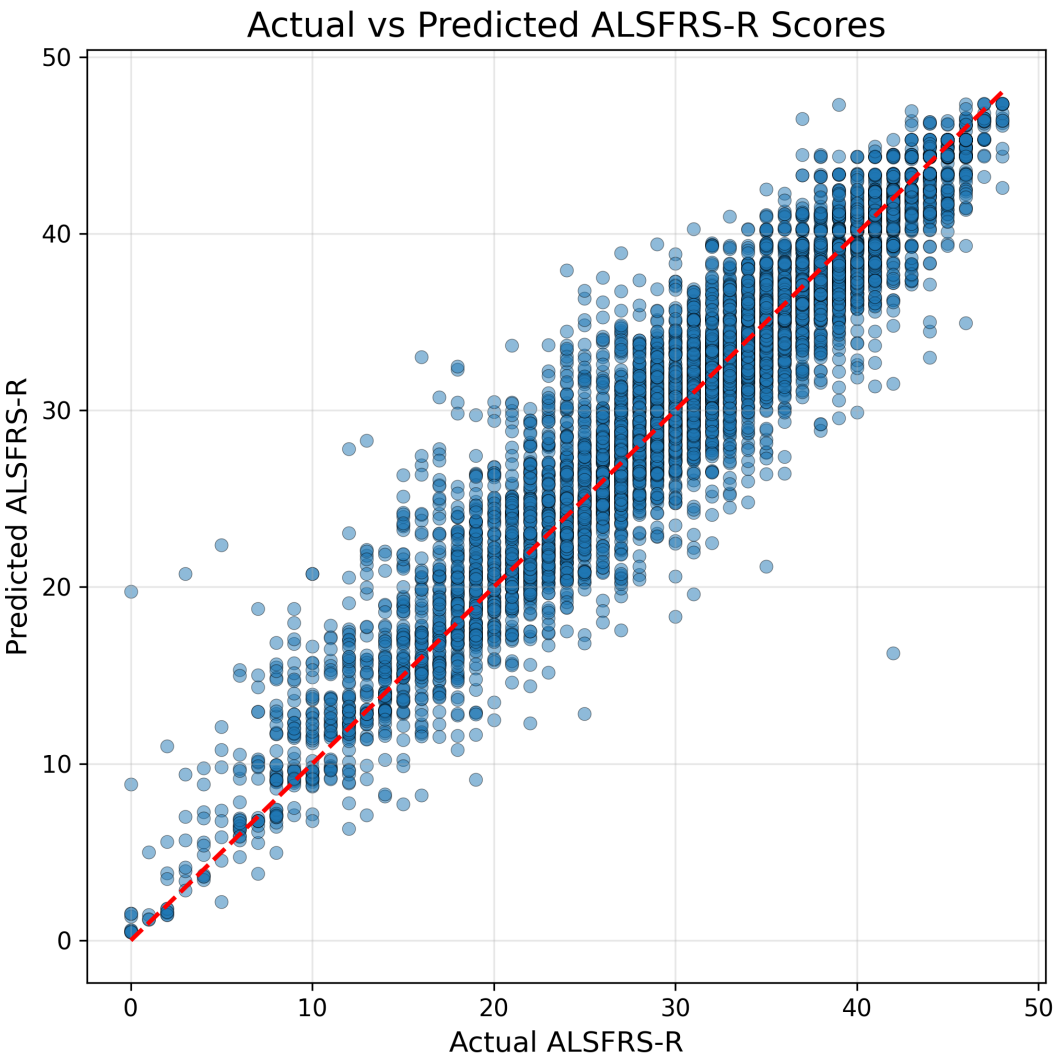

Figure 3: Actual versus predicted ALSFRS-R total scores. The dashed diagonal line represents perfect prediction. Points clustered near the line indicate that the model captures the overall relationship between longitudinal patient features and ALSFRS-R score, while spread around the line reflects patient-level prediction error.

This figure also shows why the model should be interpreted carefully. The model captures the general severity range well, but there is still spread around the diagonal. In a clinical or research setting, this means the model can provide a useful estimate of expected functional status, but it should not be treated as a certain prediction for an individual patient. A single score prediction should be interpreted alongside patient history, clinical context, and the model's expected error.

4.4 Error by Disease Severity

Figure 4 shows mean absolute error by ALSFRS-R severity group. The model had the highest error for lower-functioning patients and the lowest error for higher-functioning patients. This result is important because it shows that strong overall performance does not mean equal performance for all patients.

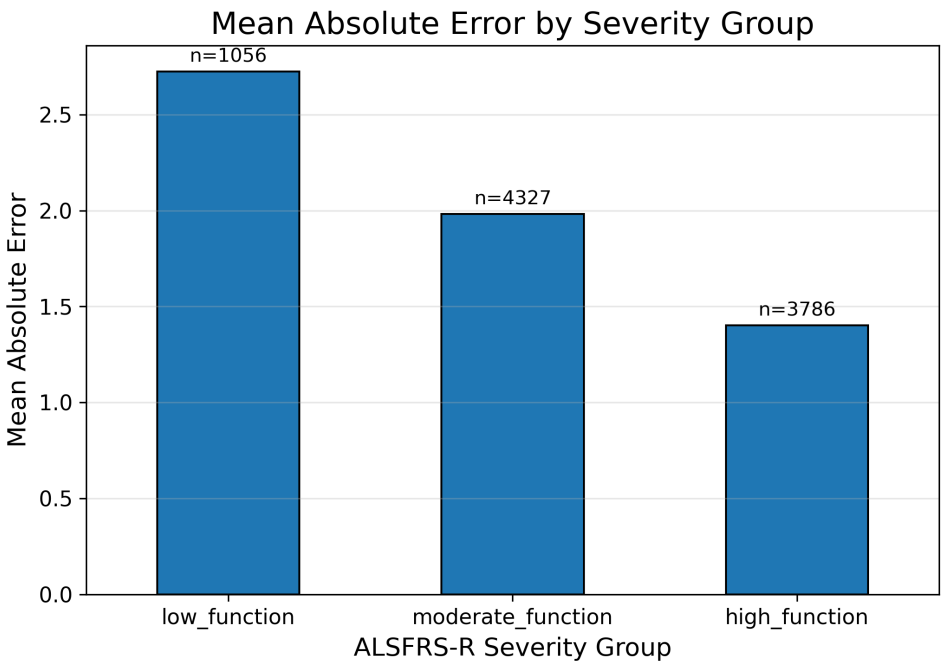

Figure 4: Mean absolute error by ALSFRS-R severity group. The model has the highest average error for lower-functioning patients and the lowest average error for higher-functioning patients, showing that prediction reliability varies by disease severity. MAE is measured in ALSFRS-R score points.

This is one of the most important findings in the project. The low-function group had an MAE near 2.7 ALSFRS-R score points, while the high-function group had an MAE closer to 1.4 points. This means the model was roughly twice as wrong on average for the lower-functioning group compared with the higher-functioning group. Since lower-functioning patients represent more advanced disease, this limitation matters in real-world use. The model appears more reliable for monitoring patients with milder impairment than for making predictions about patients with more severe functional decline.

This pattern may have several causes. Lower-functioning patients may have more nonlinear or variable progression. There may also be fewer examples of very low ALSFRS-R observations in the data, as shown earlier in Figure 2. In addition, structured visit data may not fully capture important clinical or biological factors that drive advanced-stage decline. This supports the future direction of applying the same machine learning workflow to more complete clinical, genetic, and biomarker data.

4.5 Clinical and Research Interpretation of Metrics

The strongest model achieved an MAE of about 1.81 ALSFRS-R score points. Since the ALSFRS-R scale ranges from 0 to 48, this means the model's predictions were off by about two points on average. This is useful for estimating general trajectory patterns, but it is not precise enough to replace clinical judgment. A two-point error may be acceptable for broad research modeling, but a clinician would still need to interpret the prediction in the context of the individual patient's symptoms, visit history, and treatment plan.

The RMSE of about 2.57 is higher than the MAE, which indicates that some predictions had larger errors. This is expected in clinical data because ALS progression is heterogeneous. It also supports the need for residual and subgroup analysis. Average error alone does not tell us whether the model performs consistently across all patient groups.

The R^2 value near 0.91 means the model explains about 91 percent of the variation in ALSFRS-R scores across the held-out test data. If a model had an R^2 around 0.85, that would still mean it explains most of the dataset-level variation. However, this does not mean the model is 85 percent or 91 percent accurate for a specific patient. For a clinician using this type of model with one patient, R^2 should be understood as evidence that the model captures broad population-level patterns. It should not be interpreted as certainty about an individual patient's future score.

This distinction separates a model evaluation from a usable tool. A high R^2 suggests that the model can support research tasks such as trajectory estimation, patient grouping, or identifying cases that deviate from expected progression. It does not mean the prediction should be used alone to make clinical decisions. The severity-group results make this point even clearer because the model is less reliable for lower-functioning patients. A clinician or researcher would need to treat predictions for those patients with more caution.

5 Discussion

The project showed that longitudinal clinical history is the strongest source of predictive signal for ALSFRS-R progression. The previous ALSFRS-R score was especially important because recent function is highly related to near-future function. This explains why the naive previous-visit baseline was already strong. It also explains why the learned models improved only modestly over the baseline.

Compared with the naive baseline, the learned models are more flexible because they can combine multiple features. The baseline mostly carries forward the previous score. Linear regression adds information from timing and patient-level features but assumes a linear relationship. Random forest handles nonlinear relationships and interactions, which is helpful because ALS progression is not uniform. Gradient boosting performed slightly best because it can correct errors across sequential decision trees. Even

so, the results suggest that adding more model complexity has limited benefit if the available features do not capture enough information about the patient's underlying disease process.

A major challenge was working with longitudinal healthcare data. Healthcare datasets are rarely ready for modeling without careful cleaning. Missing values, inconsistent column names, repeated measurements, and uneven visit timing all affect how the data can be used. In this project, preserving the repeated-visit structure was important because the research question focused on progression over time. The modeling dataset needed to represent patient history without accidentally removing the time dimension.

Another limitation is that the final model depends strongly on prior observed visits. This means it works best when a patient already has some longitudinal history available. For a new patient with limited prior data, the model would have less context and may be less reliable. This is important in real-world settings because patient records are often incomplete or unevenly spaced.

The model also relies mostly on structured clinical features. It does not fully incorporate richer biological signals such as biomarkers, genetic information, or molecular subtypes. Since ALS is clinically and biologically heterogeneous, structured visit history alone may not capture all factors that explain rapid decline. This limitation is especially relevant for lower-functioning patients, where error was highest.

The model should therefore be viewed as a decision-support or research tool rather than a standalone clinical tool. It may be useful for estimating general progression patterns, comparing expected and observed trajectories, or supporting patient grouping in research. It should not be used by itself to make clinical decisions for an individual patient. Future work can apply the same machine learning workflow to more complete clinical, genetic, and biomarker data. External validation on a separate ALS dataset would also help determine whether the pipeline generalizes beyond the PRO-ACT-based modeling setup.

6 Conclusion

This project built a reproducible Python workflow for predicting ALS disease progression using longitudinal ALSFRS-R data. The main research question was whether prior clinical history and patient-level features could be used to predict future ALSFRS-R total score. The results show that longitudinal clinical history supports reasonably strong prediction of ALSFRS-R score, with the best model achieving approximately 1.81 MAE, 2.57 RMSE, and an R^2 near 0.91.

The model comparison showed that learned machine learning models improved over a naive previous-visit baseline, but the improvement was modest. This means that previous ALSFRS-R score already contains substantial predictive information. Linear regression provided an interpretable benchmark, random forest captured nonlinear relationships, and gradient boosting achieved the strongest overall metrics.

The most important interpretive result was that prediction reliability varied by disease severity. The model had the highest mean absolute error for lower-functioning patients and the lowest error for higher-

functioning patients. This means the model's overall accuracy should not be interpreted as equal reliability for every patient group. In a real-world clinical or research setting, the model would be most appropriate as a decision-support or trajectory-estimation tool, not as a standalone predictor for individual patient decisions.

Overall, the project shows that longitudinal clinical history can be used to predict ALS progression with reasonable accuracy, but model performance still depends on disease severity and the amount of patient data available. Future work can apply this machine learning workflow to more complete clinical, genetic, and biomarker data.

A Appendix

This report includes the essential figures and tables needed to understand the project. Additional notebooks, generated reports, and diagnostic figures are available in the project GitHub repository: `https://github.com/ddanielkang/DATS-4001-Capstone-Project`.

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